

```

101.     return points
102.
103.
104. def rotatePoint(x, y, z, ax, ay, az):
105.     """Returns an (x, y, z) tuple of the x, y, z arguments rotated.
106.
107.     The rotation happens around the 0, 0, 0 origin by angles
108.     ax, ay, az (in radians).
109.     Directions of each axis:
110.         -y
111.         |
112.         +-- +x
113.         /
114.         +z
115.     """
116.
117.     # Rotate around x axis:
118.     rotatedX = x
119.     rotatedY = (y * math.cos(ax)) - (z * math.sin(ax))
120.     rotatedZ = (y * math.sin(ax)) + (z * math.cos(ax))
121.     x, y, z = rotatedX, rotatedY, rotatedZ
122.
123.     # Rotate around y axis:
124.     rotatedX = (z * math.sin(ay)) + (x * math.cos(ay))
125.     rotatedY = y
126.     rotatedZ = (z * math.cos(ay)) - (x * math.sin(ay))
127.     x, y, z = rotatedX, rotatedY, rotatedZ
128.
129.     # Rotate around z axis:
130.     rotatedX = (x * math.cos(az)) - (y * math.sin(az))
131.     rotatedY = (x * math.sin(az)) + (y * math.cos(az))
132.     rotatedZ = z
133.
134.     return (rotatedX, rotatedY, rotatedZ)
135.
136.
137. def adjustPoint(point):
138.     """Adjusts the 3D XYZ point to a 2D XY point fit for displaying on
139.     the screen. This resizes this 2D point by a scale of SCALEX and
140.     SCALEY, then moves the point by TRANSLATEX and TRANSLATEY."""
141.     return (int(point[X] * SCALEX + TRANSLATEX),
142.           int(point[Y] * SCALEY + TRANSLATEY))
143.
144.
145. """CUBE_CORNERS stores the XYZ coordinates of the corners of a cube.
146. The indexes for each corner in CUBE_CORNERS are marked in this diagram:
147.     0---1
148.     /|  /|
149. 2---3 |
150. | 4-|-5
151. |/  |/
152. 6---7"""
153. CUBE_CORNERS = [[-1, -1, -1], # Point 0
154.                  [ 1, -1, -1], # Point 1
155.                  [-1, -1,  1], # Point 2

```